

DATA MODELING PARADIGMS EVERY DATA ENGINEER SHOULD KNOW



BY MICHAEL FERNANDES

1. RELATIONAL MODELING (3NF)

Data is organized into normalized tables to minimize redundancy.

High Data Integrity

Reduced Duplication

Multiple Table Joins

Structured Schema

SQL-Based Access

Better Consistency

Database Normalization

1NF



- Data is stored in tabular format
- Each column holds atomic values (no lists)
- No repeating groups or arrays

2NF



- Meets 1NF
- No partial dependencies on composite primary keys
- Data split into smaller tables

3NF



- Meets 2NF
- No transitive dependencies
- Each column depends only on the primary key

2. DIMENSIONAL MODELING

Data is organized into Facts and Dimension Table used to optimize analytics and reporting

Fact and
dimension
tables

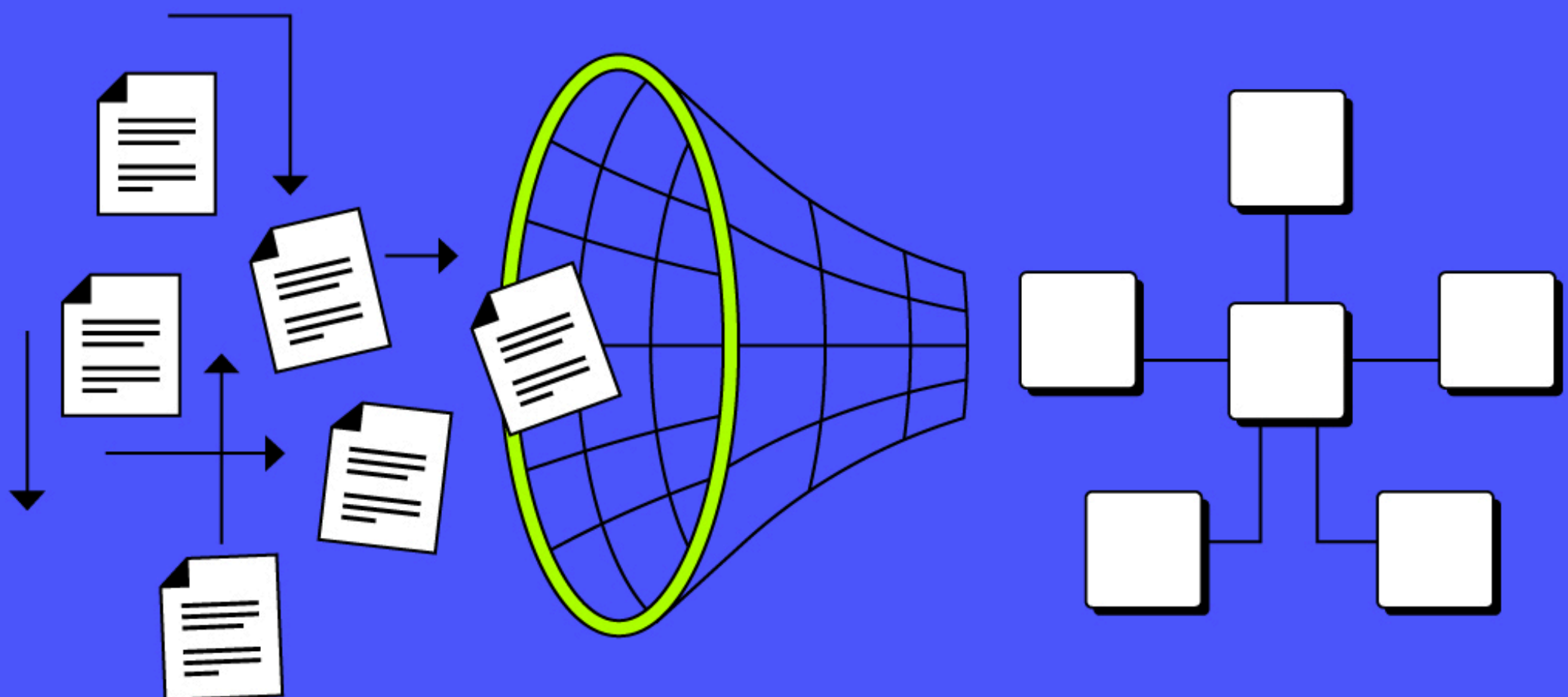
Business-
friendly
design

Optimized
for reporting

Easy to
understand

Supports
historical
analysis

High
analytical
performane



3. STAR SCHEMA

A central fact table connected directly to dimension tables.

Central fact table

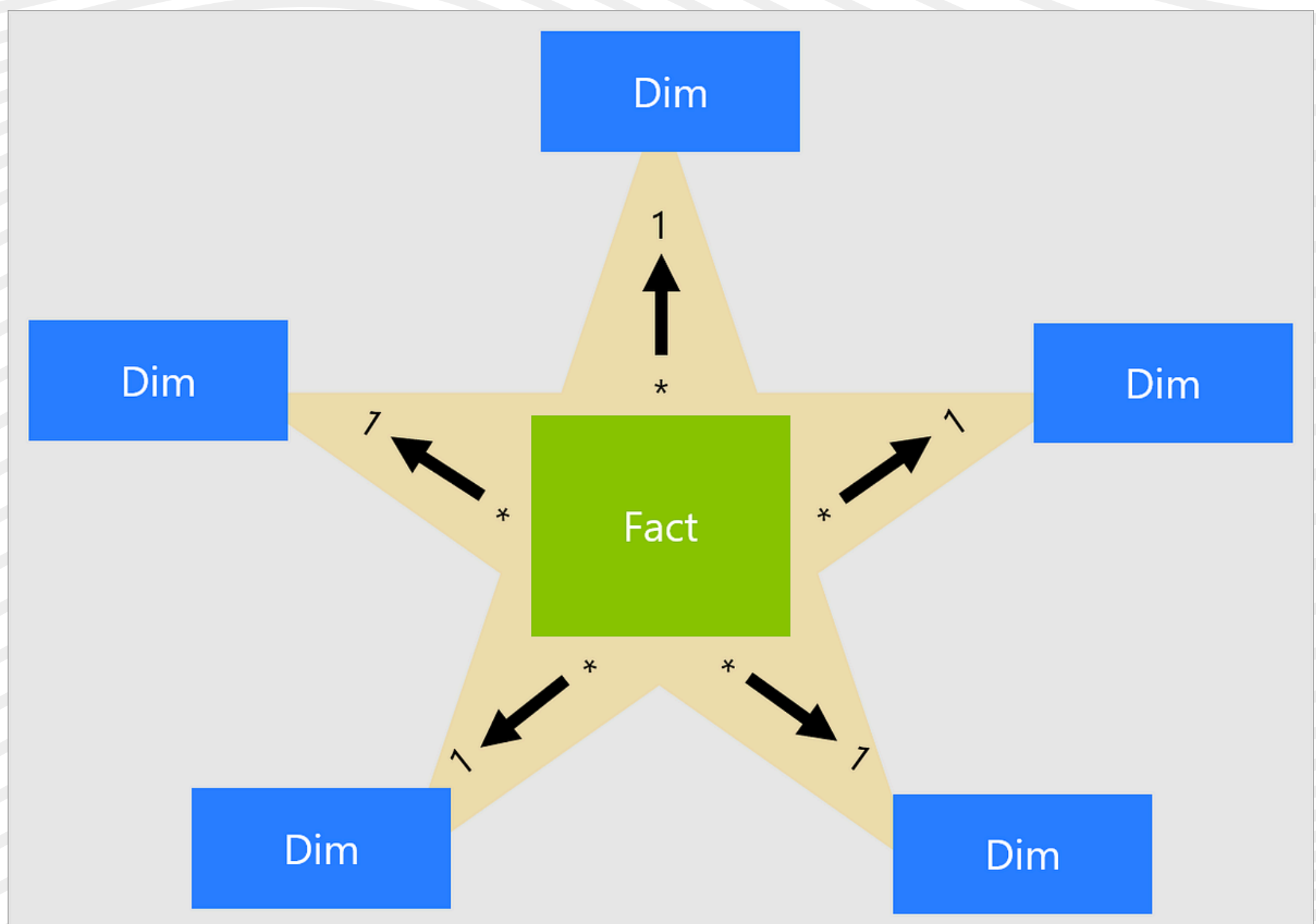
Denormalized dimensions

Simple query structure

Fewer joins

Fast aggregations

Excellent BI performance



4. SNOWFLAKE SCHEMA

Dimension tables are further normalized.

Normalized
dimensions

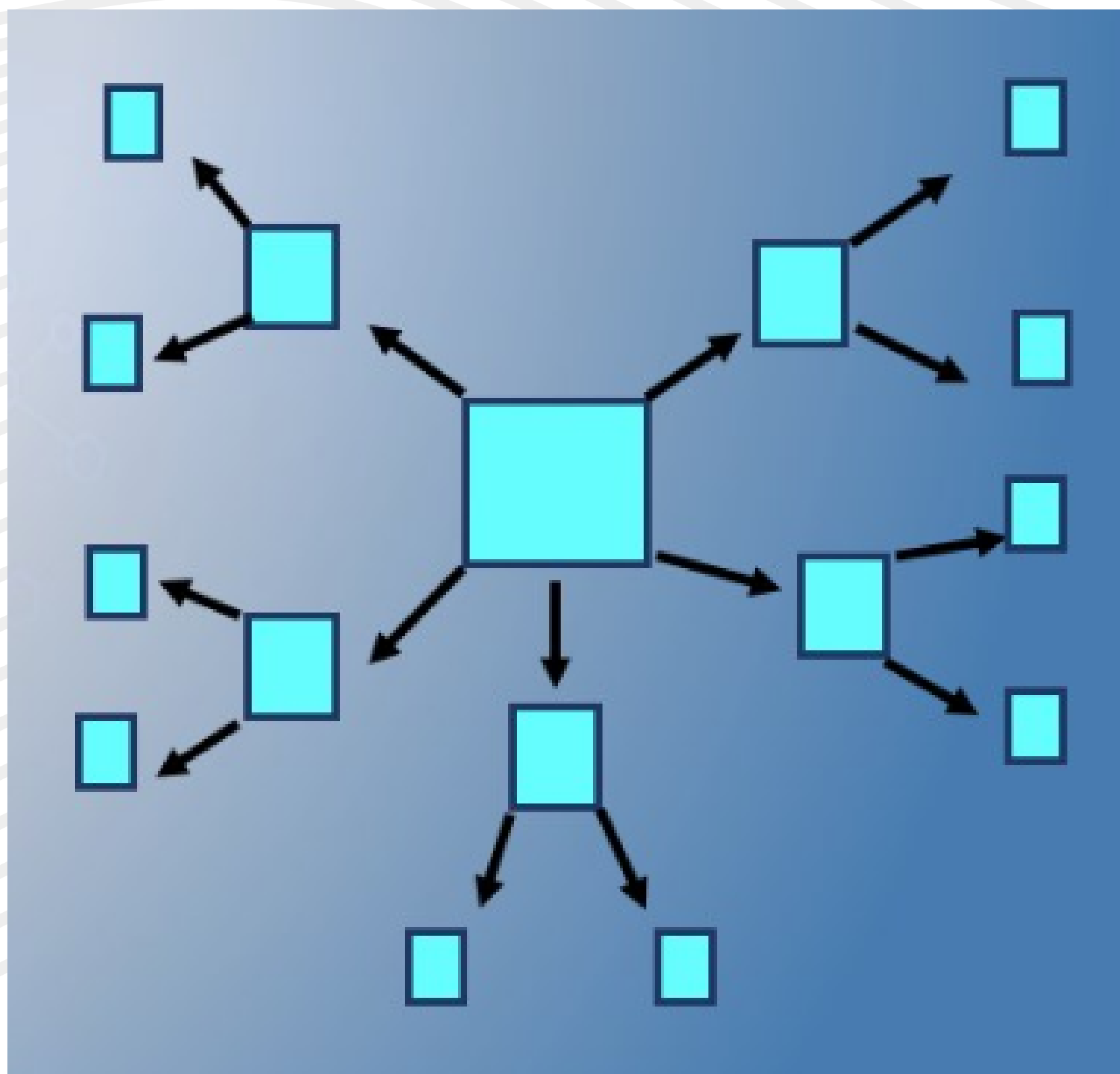
Less
redundancy

Additional joins
required

Reduced
storage usage

More complex
design

Suitable for large
dimensions



5. DATA VAULT MODELING

Designed for scalable enterprise data warehouses.
Contains Hubs, links and Satellites

Hubs store
business keys

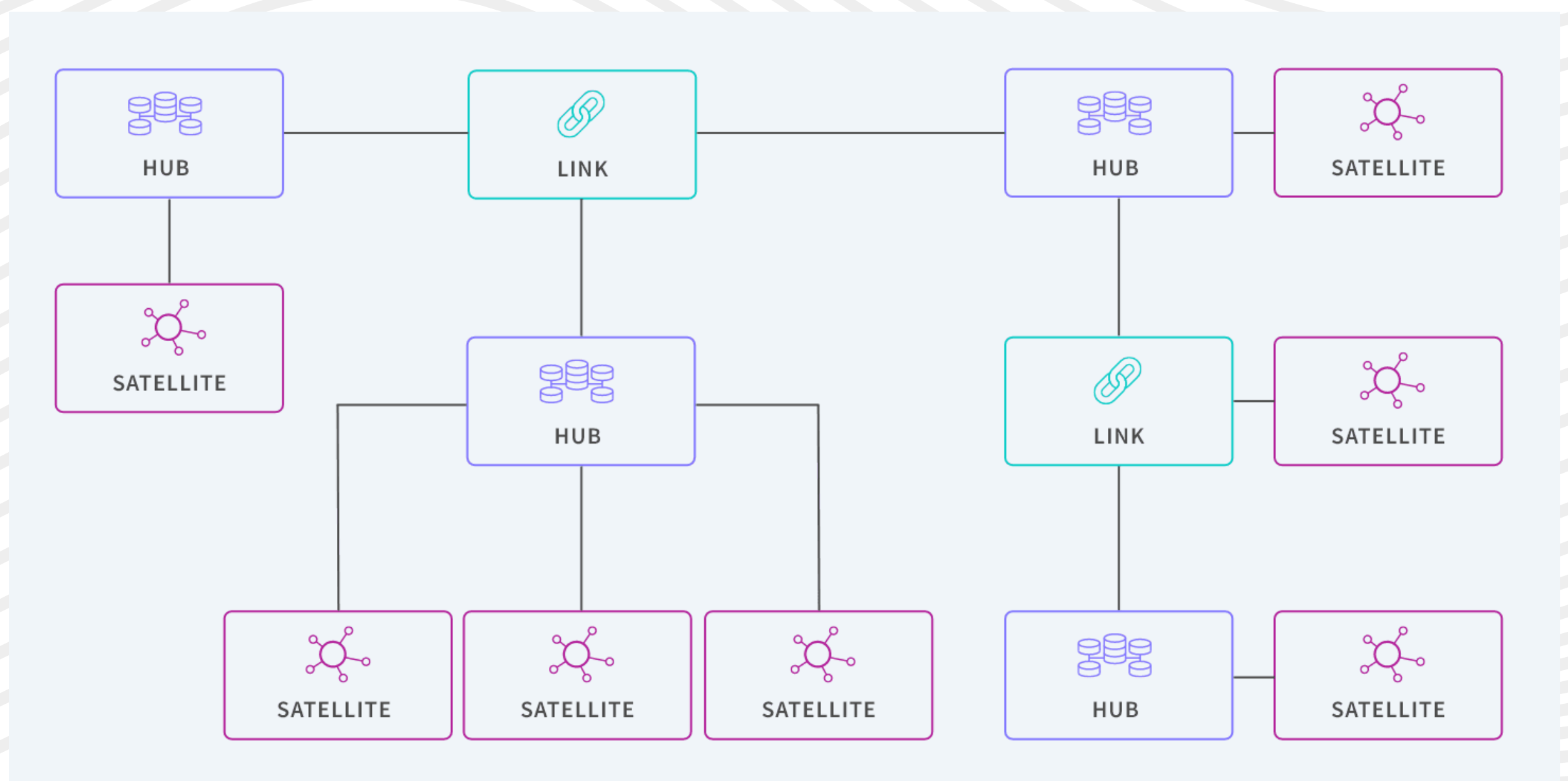
Links store
relationships

Satellites store
attributes

Complete
historical
tracking

Highly
scalable
architecture

Supports
agile
developmen



6. ONE BIG TABLE (OBT)

All data stored in a single denormalized table.

Fully
denormalized

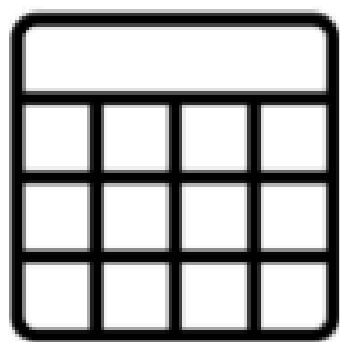
Minimal
joins

Extremely
fast queries

Easy for
analysts

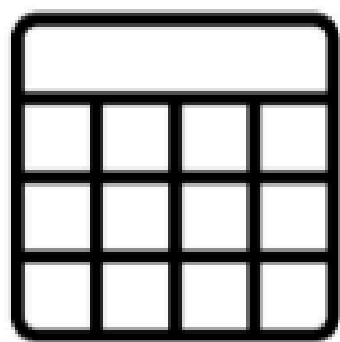
Higher
storage
consumptin

Common in
cloud
warehouses

A small 4x4 grid representing a table with 4 columns and 4 rows of data.

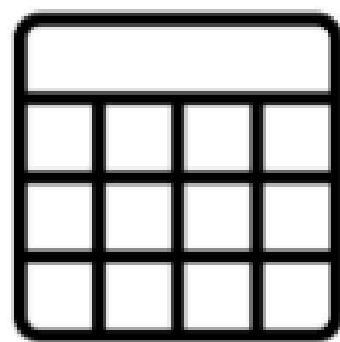
Movie

x

A small 4x4 grid representing a table with 4 columns and 4 rows of data.

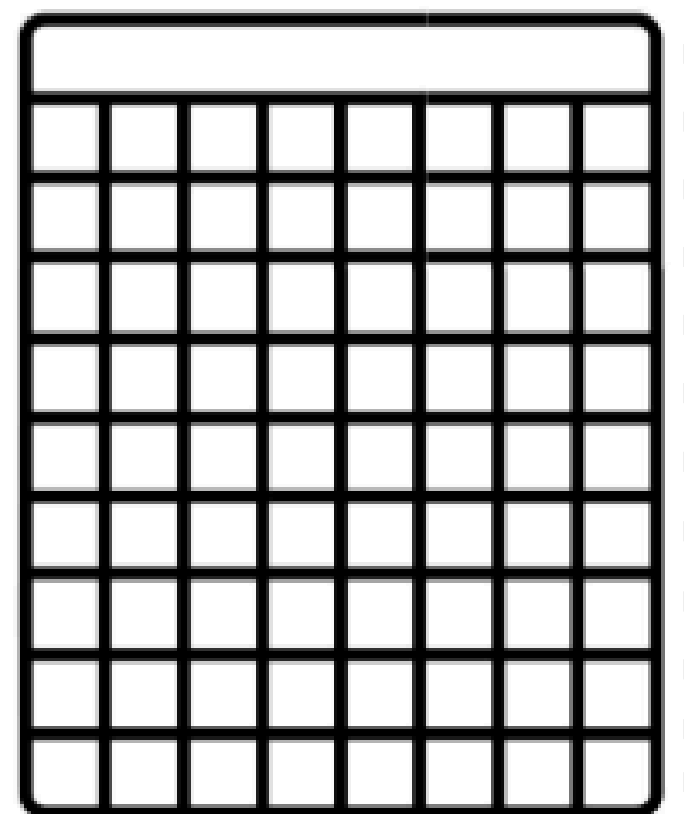
Customer

x

A small 4x4 grid representing a table with 4 columns and 4 rows of data.

Rental

=

A large 12x12 grid representing a single denormalized table containing all data from the three smaller tables.

OBT

7. LAKEHOUSE MODELING

Combines Data Lake flexibility with Data Warehouse structure.

Medallian Architecture

Supports structured and unstructured data

Data quality improves layer by layer

ACID transactions supported

Highly scalable

Enables analytics and ML together



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8. ANCHOR MODELING

Highly flexible model designed for frequent schema changes.

Schema evolution friendly

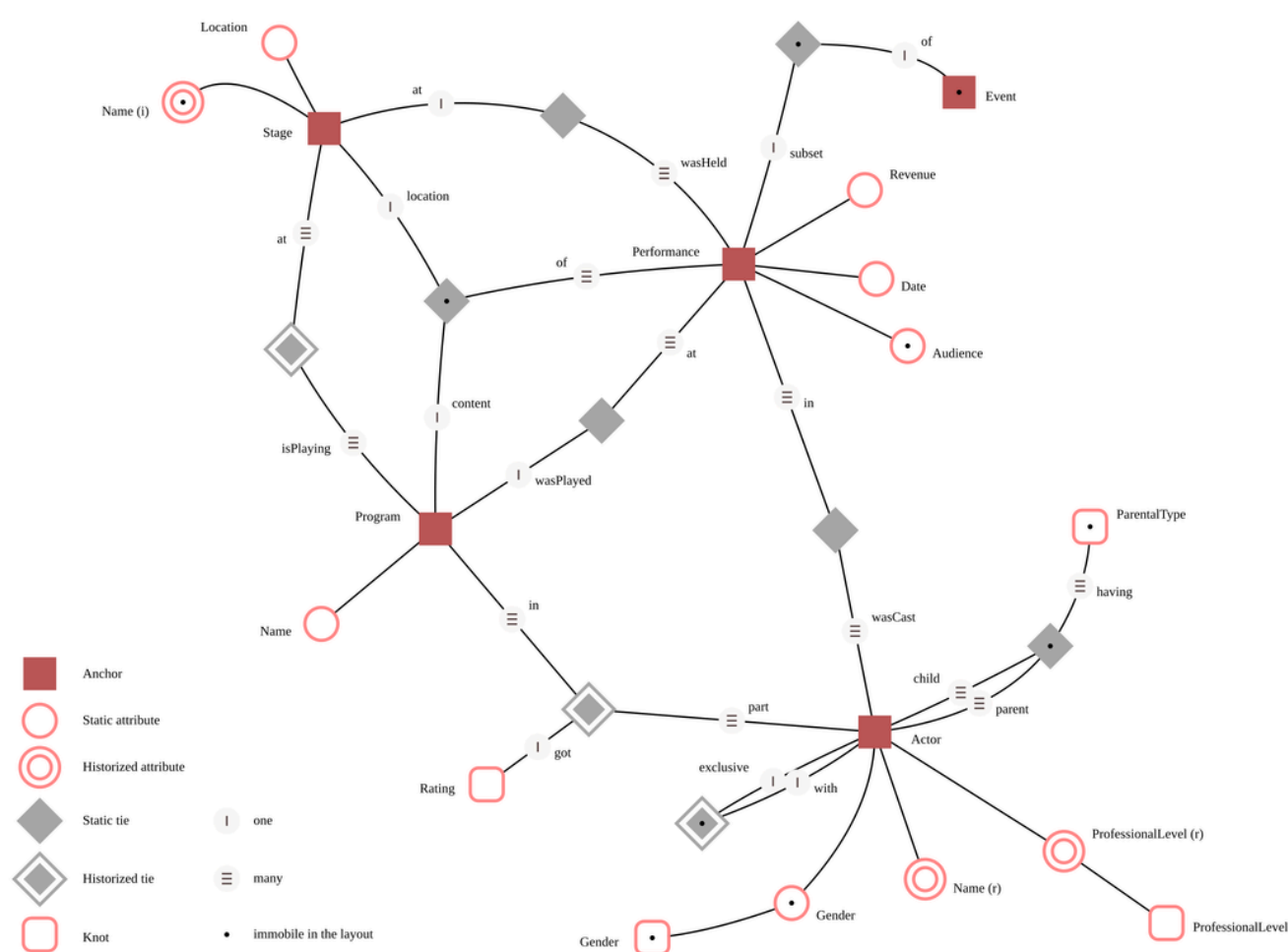
Highly normalized

Historical tracking

Flexible architecture

Supports changing business

Suitable for enterprise systems



9. GRAPH MODELING

Stores relationships as nodes and edges.

Node-edge
structure

Relationship
-centric

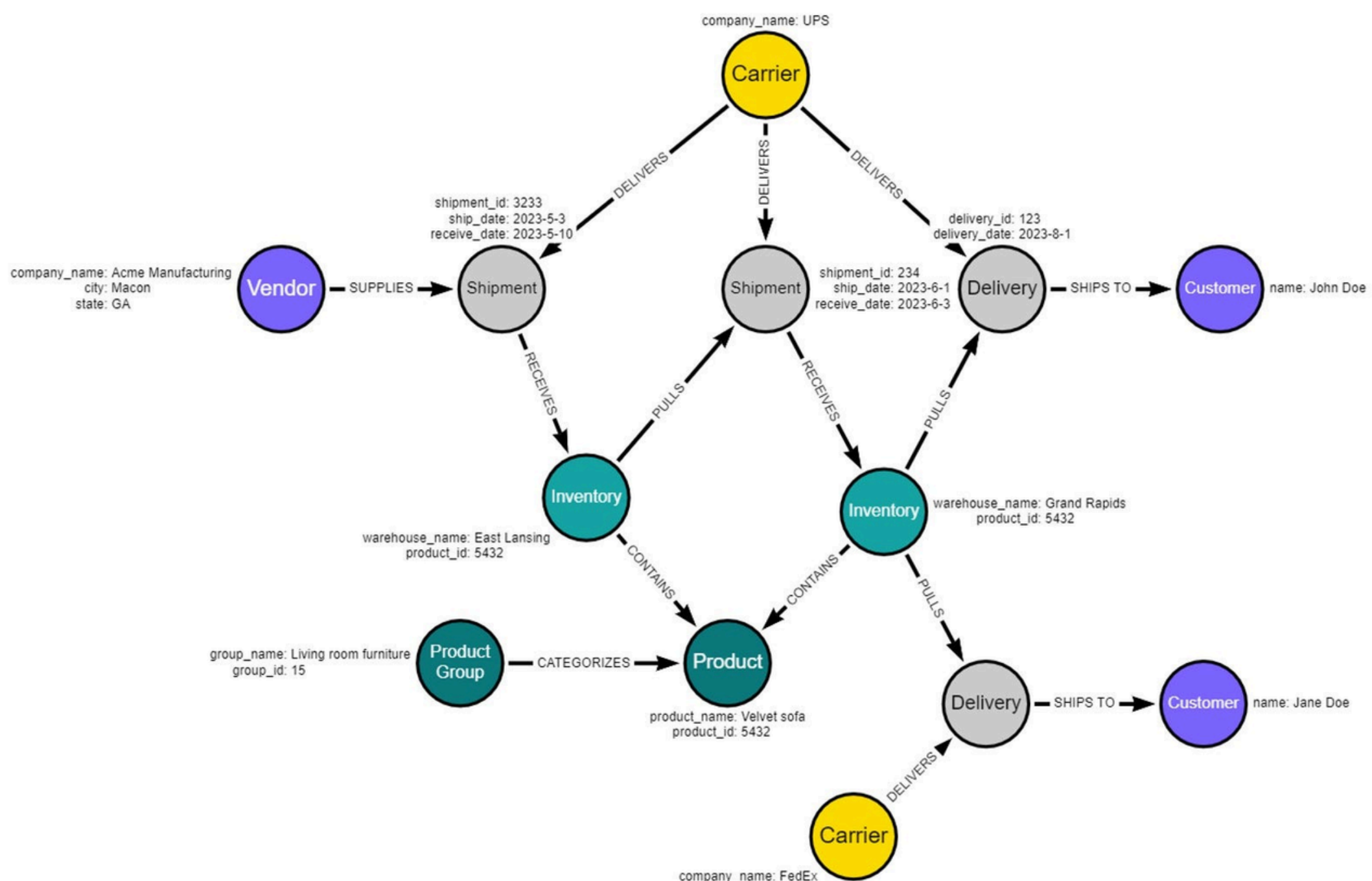
Efficient
graph
traversals

Flexible
schema

Handles
connected
data

Excellent for
network
analysis

Graph Model: Supply Chain



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10. DOCUMENT MODELING

Stores data as JSON-like format called documents.
Has Key-value pair

JSON-based
storage

Flexible
schema

Supports
nested data

High write
performance

Easy
horizontal
scaling

Ideal for
semi-
structured

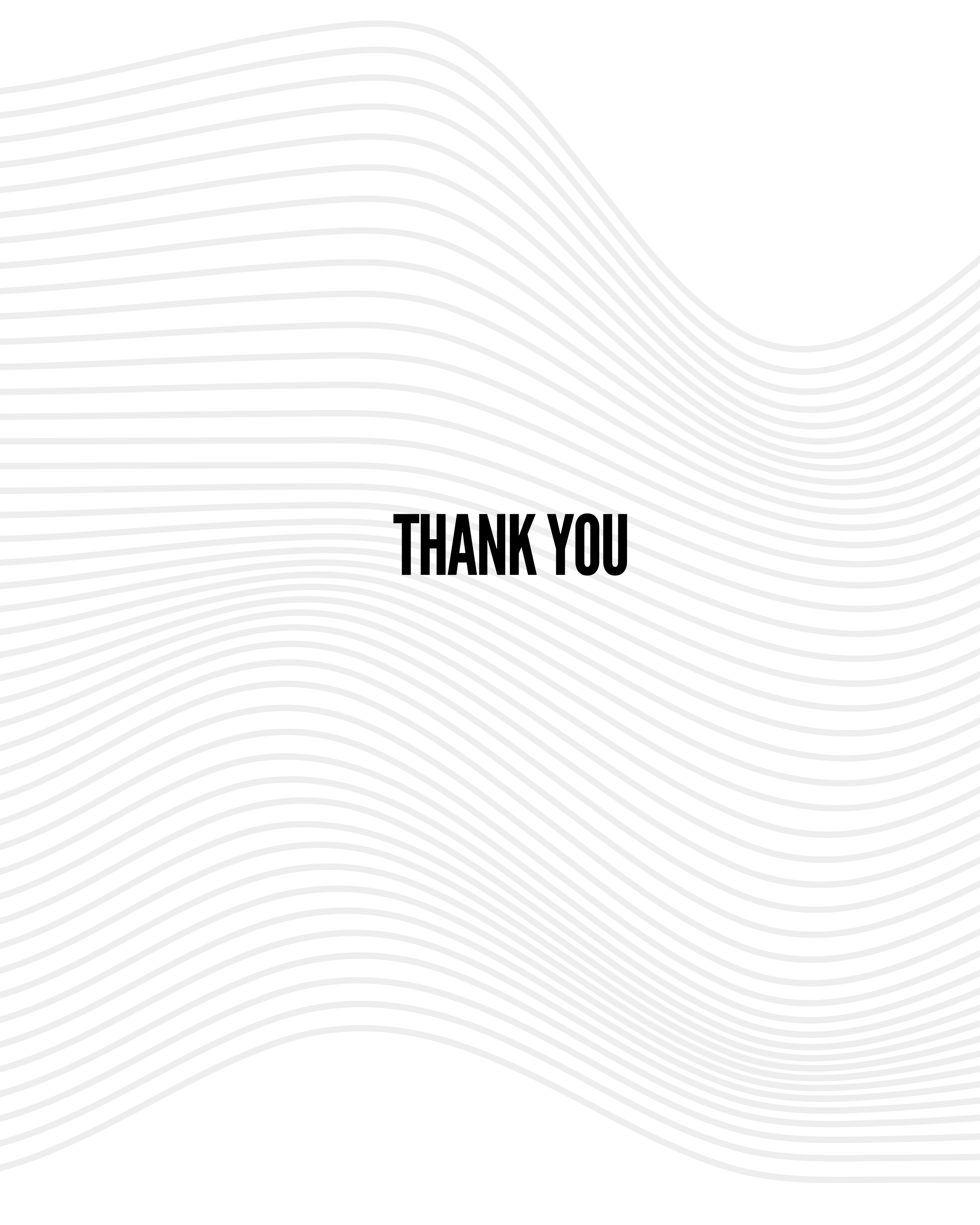
MongoDB Document Model

RDBMS



MongoDB



The background of the image consists of numerous thin, light gray lines that flow in a wavy, undulating pattern across the entire frame. These lines create a sense of movement and depth, resembling a stylized topographical map or a liquid surface. The lines are more densely packed in some areas and more spread out in others, contributing to the overall organic feel of the design.

THANK YOU